Bassez 2021 Breast Cancer — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

θ=10 Harmony on sample: iLISI 4.9→8.6/41 (p10 3.2). θ-sweep on the embedding: θ=2 fragments (45% main), θ=10 one continuous body (>99.9%). Lobes span all 41 patients (max 0.17). Complete island removal (0.17% specks). GO.

Gate 2 — module geography → GO (non-branching)

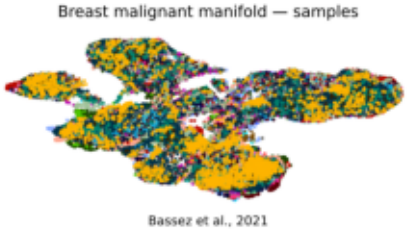
Gast~pSom co-vary (raw −0.10 / sm −0.21), both oppose Crest (sm −0.24 / −0.82). Along pseudotime Gast/pSom fall (ρ−0.12/−0.27), Crest rises (ρ+0.30); joint pSom&Crest 0.23<1 — inverse terminal gradient. One axis, no arms. No inversion.

Gate 3 — hub cross-repression → NA

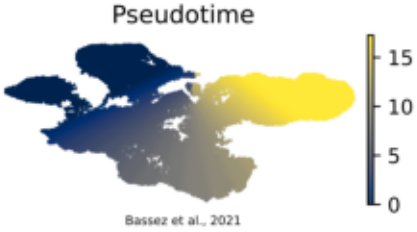
Non-branching (curve 2 tips/0 forks; PPT 0 Gast-enriched tips). No bifurcation → NA.

STEP FIGURES

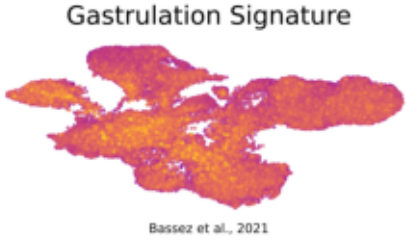
Manifold — samples



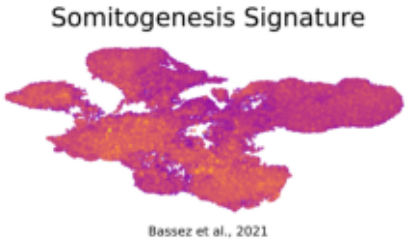
Pseudotime



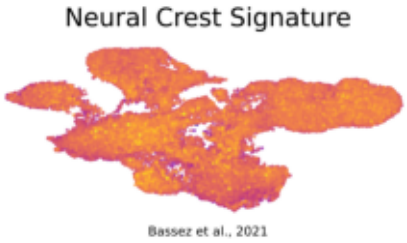
Gastrulation



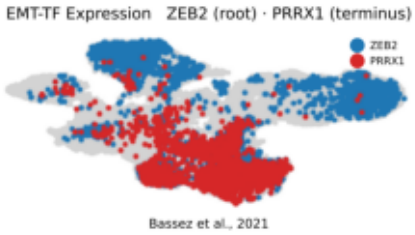
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (Bassez et al., 2021), 10x — 65,885 malignant cells; 65,615 analysed (41 samples, ≥20 cells).

Verdict.
The embryonic pattern reproduces as one continuous (Gast/pSom) → Crest axis — a linear (non-branching) manifold. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs ZEB2 (early/root) → PRRX1 (late/terminus).

Parameter choices

Threshold	20 cells/sample → 41/42 kept (gap-finding; anchor 100 drops 39 real samples)	**Embedding**	n_pcs=30 (scree, centres 20–40); Harmony θ=10 (iLISI plateau + continuous embedding)
Islands	single-pass radius-graph specks (0.17%); no re-embedding	**Trajectory**	scFates curve, N=20 (2 tips, 0 forks, fold-free self-approach 2.6)
Root	tip 2 = Gast/pSom-high, Crest-low (Crest is late, never root)	**Geometry**	non-branching / linear (Gast/pSom → Crest single axis; corr x_lin~t = +0.94)
EMT-TFs	ZEB2 (root) → PRRX1 (terminus), detection-gated		